Retrieval-Augmented Generation (RAG)

Introduction

Retrieval-Augmented Generation combines information retrieval with large language models. It allows a model to use relevant external documents when answering questions.

How RAG Works

1. Documents are loaded and divided into smaller chunks.
2. Text chunks are converted into numerical embeddings.
3. Embeddings are stored in a vector database.
4. A user submits a question.
5. The retriever finds relevant document chunks.
6. The language model generates an answer using retrieved context.

Benefits

RAG can reduce unsupported answers, use information from external documents, improve relevance, and support knowledge-intensive applications.

Challenges

Poor document quality, unsuitable chunk sizes, and irrelevant retrieval can reduce answer quality. Language models can still generate claims that are not supported by the retrieved context.

Evaluation Metrics

Faithfulness: whether an answer is supported by retrieved context.
Answer relevancy: whether the answer addresses the question.
Context precision: whether relevant information is ranked highly.
Context recall: whether retrieved context contains the information needed to answer.

Conclusion

A reliable RAG system requires good document processing, effective retrieval, accurate generation, and systematic evaluation.